

Machine Learning for Football Matches and Tournaments

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Abstract— The excitement of predicting the result of a football game and support for their favourite clubs has brought all the sports enthusiasts together to develop platforms around it. Given the access to data, football analytics has emerged as a huge industry not limited to the players and teams. This research manipulates data available for different football leagues and concepts of classification models to predict the **result of a match at half time with 80% accuracy** (using **ensemble techniques**) for a match being played in the English Premier League and the winner of the same league **using the historical and current performance of the club and the players in the league at a match week level with up to 95% accuracy**. The research uses a dynamic dataset and proposes an application that requires minimum user input for prediction. Further, simple features and visualization of results make the research understandable and reusable.

Keywords— classification, league winner, match week, data preparation, winning probability.

I. INTRODUCTION

Football is a global sport, played and watched in every part of the world. Different national and international leagues, divisions, tournaments, etc., with different competition structures and fixtures. Generally, there is more than one football tournament going on at a time, and fans and football enthusiasts are eager to watch the game's results uncover. Many virtual platforms have also been developed around football with a whole community, such as FIFA, Fantasy Premier League, etc. Given that several organizations manage these tournaments offline and online, there is also a huge database for every action of every game. The data is present in the forms of numbers, images, videos, etc., at different levels, i.e., player level, match level, league level, playing ground level, coach level, country level, etc.

This research work is a continuation of the done by the authors in the paper 'Full-Time Result

Prediction using ensemble techniques' as in [1]. We have taken advantage of the freely and easily accessible data for one of the biggest leagues in Europe, i.e., the English Premier League or the Premier League, for our research. The understanding of data clubbed with the understanding of the game has helped us formulate a way of predicting the result of a football match at halftime and the winner of the league/tournament based on their historical and current performance. We have divided the research into two parts: match winner prediction and league winner prediction.

The first part of the research establishes how adding features to the data can help predict the results of a game better. We show the effect of added features and compare it with previously computed results. For the second part, we shift our focus from the match level to the league level, i.e., we do not consider two teams in the prediction model but only one (each of the participating clubs in the league). We perform a set of operations to convert the data used in the first part for the latter part. We use our understanding of the game to get data at a match week level and then predict the league's winner on a team and match week level. We further develop an application to make our research accessible to everyone and use visual tools to display the modeling results.

In the first section of the paper, we have described the frequently used terms in the later sections of the paper and detailed similar work done in the past. The next section discusses the first part of the research, the dataset, features, modeling, results, and comparison. The latter part of the research is described, along with the preprocessing

and modeling steps. In the last section of the paper, we have added glimpses of our flask application and the output. We discuss the applications and further scope of the research in the end.

II. BACKGROUND

We had previously defined home team and away team based on the stadium the match is being played. Each team has 11 players, and a game goes on for 90 minutes with two halves of 45 minutes each. A league is a tournament in which different teams play against each other a defined number of times, and the team with the highest points is declared the league winner. In the second part of the research, we repeatedly talk about match week. Match week, generally 7 days, is a stipulated time during the season each participating team has a match. The English Premier League, whose data the research and modeling are being done, has 38 match weeks with 10 matches each week, i.e., 380 matches in total every season. Twenty teams participate in the league, and only one is declared the winner.

Other terms used in the research are around the dataset's features defined in the first part of the research. Here, we consider leader board standings, which is the team's rank that is secured in the last season if participated. The form is used to define the playing form of the team, that is, whether it has been losing or winning the previous matches played by it as a home or away team in the league. Form points are calculated similarly to how a team's points are calculated: +3 for the win, 1 for the draw, and 0 for loss. We also introduce terms like goals scored and goals conceded which are the goals secured by the playing team and the goals scored by the opponent team in a match cumulated over the match weeks.

Other common terms used in the research are data preparation, pre-processing, modelling, testing, accuracy, cross-validation score, visualization, classification, etc., which have their usual meanings.

A. Literature Survey

Sports analytics as a field has interested many researchers and statisticians. In the past, many

efforts have been made to predict the results of a football match and have been successful too. Given that there are so many different leagues played in different countries, the research is not limited to any one kind of football tournament or match. The motivation of this paper sparked from similar research in cricket described in the paper 'Using Machine Learning to Predict the Outcome of English County twenty over Cricket Matches' by Stylianos Kampakis as in [2]. Statistics played an important role in understanding the nature of data. The work done by David Sheehan in 'Predicting Football Results with Statistical Modelling' helped us analyze important features of our data. This and the previous work on which it is based is very deeply influenced by the work done by Rudraksh Tuwani and Siraj Raval using the English Premier League data. The former, worked on data preparation and analysis, whereas the latter, built a basic predictive model for the same and discussed its scope. However, all the work done and quoted in the author's previous work was based on predicting the results of a match and not a tournament or league. Ryan Beal, Stuart E. Middleton, Timothy J. Norman and Sarvapali D. Ramchurn described using a week-by-week analysis in their work 'Combining Machine Learning and Human Experts to Predict Match outcomes in Football: A Baseline Model' as in [3].

In their research, 'Predicting English Premier League Winners using Machine Learning', Divyesh Harit and Rishi Mody used various algorithms on a long format data till 2014 and predicted the winner with ~60% accuracy. They detailed the challenges of getting data in the right format, which we have tried to overcome in this research. There also has been much traction towards fantasy football, whose features we have used in our research. One such work 'Deep Artificial Intelligence for Fantasy Football Language Understanding' by Aaron Bughman, Micah Forester and Jeff Powell widened our horizon about the applications of the concepts using Machine Learning and Deep Learning as in [4]. We further explored how we could include player data for predicting results of matches and leagues inspired by other work like PlayerRank as in [5].

III. MATCH WINNER PREDICTION

This part of the research extends previous research work in 'Full-Time Result Prediction using ensemble techniques' as in [1]. Here, we try to predict the result of a football match played in the English Premier League at halftime in terms of the winning probability of the home team based on the goals scored till halftime, the current leaderboard position of the teams in the league, previous year leader board standing and the performances in the earlier matches. In comparison to the previous research, we have added independent features such as the form, form points, etc., and considered goals scored and conceded by the teams in the current season.

A. Dataset

For this research, we are using the English Premier League or the Premier League data available freely on the Football-Data.co.uk website and many others. The data's granularity is at every match level; thus, a dataset of every year considers 380 matches. The dataset consists of 45 features: match date, home team, away team, referee, goals scored at half time and full time, shots, shots on target, corners, fouls, offsides, cards, and many more. We wanted to have a prediction model based on historical and current season statistics. For this purpose, we used the premier league data from 2005 to 2020 for training (6060 matches), the last 20 matches of the 2020-2021 league as validation, and the ongoing season 2021-22 as the test set.

The dataset for each year includes five categorical variables, one ID variable, one date variable, and the rest are numerical variables. For the sake of simplicity and to make the end application easily usable by all kinds of users and not just football fanatics, we stuck to using common features for our model and then building functions on it rather than giving all the features available in the dataset to keep the number of data points high. The chosen features are names of the playing teams, goals scored by them at halftime, and we retrieve the leaderboard position of the previous year from a static self-concatenated dataset, a sample of which is shown in Fig. 1. Our dependent variable is the full-time result.

Team	2021	2020	2019	2018	2017	2016	2015	2014	2013	2012	2011
Arsenal	8	8	5	6	5	2	3	4	4	3	4
Aston Villa	11	17				20	17	15	15	16	9
Birmingham											18
Blackburn										19	15
Blackpool										18	14
Bolton											
Bournemouth		18	14	12	9	16					
Brighton	16	15	17	15							
Bradford											
Burnley	17	10	15	7	16		19				
Cardiff								20			
Charlton											
Chelsea	4	4	3	5	1	10	1	3	3	6	2
Coventry											
Crystal Palace	14	14	12	11	14	15	10	11			
Derby											
Everton	10	12	8	8	7	11	11	5	6	7	7
Fulham	18		19					19	12	9	8
Huddersfield			20	16							
Hull							18	16			
Ipswich											
Leeds	9										

Fig. 1 EPL Previous Year Standings

Here, for each team (40 in total), the final leader board position (between 1 to 20, both included) of the previous 18 years is specified. For a match being played in the current season, the leader board position of last year, i.e., 2021, would be used. The blank cells suggest that the given team did not participate in the league last season and would be filled as 0 for modeling. In the next section, we have described how multiple features were derived from these basic 7 features, i.e., Date, Home Team, Away Team, Halftime ad full-time goals for both teams, and the full-time result.

B. Features Added

All our added features in this research are arranged by teams and match week, i.e., for a match played in the third match week (third match of the season for both teams), we would consider the cumulative of all features of the previous two matches played with different or same teams. We list down the added features and their description in Table I.

TABLE I
FEATURES ADDED

Feature	Acronym	Description
Goals Scored	HTGS/ATGS	The total number of goals scored by the team in the season
Goals Conceded	HTGC/ATGC	The total number of goals conceded, by the team in the season
Cumulative points	HTP/ATP	The sum of points of each team in the season (3 for win, 1 for draw, and 0 for losing a match)
Form	HTFormPtsStr/ ATFormPtsStr	A character string consisting of W, D, or L based on results of the last 5 matches played by each team in the season
Form Points	HTFormPts/ ATFormPts	The sum of points scored by each team in the previous 5 matches

Leader board Position	HomeTeamLP/ AwayTeamLP	The final leader board position of the two playing teams
Match Week	MW	A number between 1 and 38, both included, specifying the playing week for every match
3-win streak	HTWinStreak3/ ATWinStreak3	Specifies if the team has won the previous 3 matches played in the season
5-win streak	HTWinStreak5/ ATWinStreak5	Specifies if the team has won the previous 5 matches played in the season
3 lose streak	HTLossStreak3/ ATWinStreak3	Specifies if the team has lost the previous 3 matches played in the season
5 lose streak	HTLossStreak5/ ATWinStreak5	Specifies if the team has lost the previous 5 matches played in the season
Goal Difference	HTGD/ATGD	The difference between the Goals Scored and Goals conceded by playing teams
Points Difference	DiffPts	Difference in the cumulative points of the two playing teams
Form Points Difference	DiffFormPts	The difference in the form points of the two playing teams
Leader board Position Difference	DiffLP	Difference between the previous year's standings of the two playing teams

^a. Note: HT stands for Home Team, and AT stands for Away Team.

For Goals Scored and Goals Conceded, the value is 0 for the first match played by each team as this feature consists of an aggregation of goals scored up till that point in the season. For calculating the form points, we have used the data of the **previous 5 matches, which can be increased or decreased for accuracy**. For the first five-match weeks, we add 0 points for the matches that have not been played. For example, the result of the last 5th match of a team in a season would not be available in the fifth match week. The four streak features are categorical features added to give the model more to learn about a team's performance in the current season.

Some of our different features, namely the goal difference, points difference, and form points difference, **will increase linearly as the season progresses. To abstain from bias in the model, we scale these features according to their match week**. This concluded the data preparation of our training and validation set, and next, we followed some pre-

modelling steps before fitting the model as described below.

C. Modelling

As a part of pre-modelling, we **filter the data to keep only those matches that were played third match week onwards because of the insignificant contribution of the initial matches to the team's future performance**. After separating the data into the feature set and the target variable, the numerical data were scaled to make the data comprehensible by the model and free from bias. The next pre-processing step included charting out dummies for all the categorical variables in the feature set, mainly Home Team and Away Team. For evaluation, we have split the dataset into training and validation sets in **sequential order and not random as by default**.

To notice the behaviour of different state-of-the-art classification algorithms on our dataset, we fitted and evaluated our data on 9 algorithms: XGB Classifier, Logistic Regression, Random Forest Classifier, Decision Tree Classifier, K Neighbours Classifier, SVC, Gaussian NB, Ada Boost Classifier, and Gradient Boosting Classifier. We also considered their cross-validation scores to choose the top-performing algorithms. The results of the iterations are shown in Table II.

TABLE II

COMPARISON OF ACCURACY (VALIDATION SET)

Model Name	Accuracy Score	Cross Val Score
XBG Classifier	0.75	0.637276
Logistic Regression	0.75	0.628674
Random Forest Classifier	0.70	0.624373
Decision Tree Classifier	0.65	0.539964
K Neighbors Classifier	0.60	0.564337
SVC	0.65	0.635842
GaussianNB	0.50	0.463620
Ada Boost Classifier	0.70	0.627240
Gradient Boosting Classifier	0.70	0.636380

We can see that XGB Classifier, Logistic Regression, Ada Boost Classifier, and Gradient Boosting Classifier give a similar accuracy of 70% on the validation set. Using the Cross-Validation Score, we can discard Ada Boost Classifier. The next steps included combining the top 3 algorithms

using a **Voting Classifier** to get the best results possible. Thus, for this purpose, we ensembled the three, fitted the data to our ensemble model, and tested it on the validation set for evaluation. Our ensemble model resulted in an **accuracy of 80% on our validation set.**

D. Comparison

In the research, Full-Time Result Prediction using Ensemble Techniques just like 'Heterogeneous ensembles of classifiers in predicting Bundesliga football results' [6] by Jan Kozak and Szymon Głowania, we had tried different combinations of the top-performing algorithms. We evaluated them on the same dataset with fewer features (only goals scored at halftime, points of playing teams before a particular match, and the last year's leader board position). The ensemble of 3 algorithms gave the best results: XGBoost, Gradient Boosting Classifier, and Logistic Regression, i.e., 75% accuracy as shown in table 3 and 4. A similar combination could classify the results with an 80% accuracy this time with the help of the addition of features mentioned in Table 1. An interesting point to note here is that the combination of only two classifiers, Gradient Boosting Classifier and Logistic Regression, also resulted in similar accuracy of 80% when fitted to the latter.

IV. LEAGUE WINNER PREDICTION

Building on the last part of the research, this work focuses on predicting the winner of the English Premier League as in 'Prediction Model of Football World Cup Championship Based on Machine Learning and Mobile Algorithm' [7] by Yanyang Bai and Xuesheng Zhang in terms of probability at the end of each match week. The research aims to build a predictive system capable of comparing the probability of all playing teams winning the league at the end of every match week and the trend of winning probability of each team throughout the season. For this purpose, we have used the same dataset as used in the first part of the research. We are going to use the current season, i.e., 2021-22, as our testing dataset, and for this purpose, we would require a continuous population of the dataset as the season progresses. The training

data remains the same from season 2005 to 2020. To predict the results at the match week level, we would require bringing it to a week level instead of the current match one. This is described in the sections below.

A. Pre-processing

For this application, we would be working with the following features: year of the league, match week, team name, cumulative goals scored, cumulative goals conceded, points, and form points at a match week level. The teams generally play half of the matches on home ground and the other half as an away team. So, we have considered all the matches played by the team irrespective of the ground. We concatenate the home match and away match statistics for the same, remove duplicates (if any), and then group the data on a Year and Match week level for each playing team resulting in a dataset of shape (11015, 46), including dummy variables for each team. As our dependent feature is the result of the English Premier League, we have to include it in our training set. So, a categorical variable 'Win' was included depending on the league winners, as mentioned in Table III.

TABLE III

WINNERS OF EPL (2005-20)

Year	Team
2005	Chelsea
2006	Man Utd
2007	Man Utd
2008	Man Utd
2009	Chelsea
2010	Man Utd
2011	Man City
2012	Man Utd
2013	Man City
2014	Man Utd
2015	Leicester
2016	Chelsea
2017	Man City
2018	Man City
2019	Liverpool
2020	Man City

Our goal is to predict the winner of the ongoing EPL season, i.e., 2021-22. Therefore, we would prepare the test data similar to the training data.

B. Testing Data

As the league for which the prediction is to be done would always be in progress of using the application, there is a need for recent data that would keep updating. To keep the process simple, we are loading the dataset directly from the URL. The data has 105 features, but to match our independent variables with training data, we would need only the names of the playing teams, goals scored by them at full time, and the full-time result. As the recent season was interrupted by Covid-19, the fixtures are different from the past years, i.e., all 10 matches of a match week are not played continuously. So, we calculate some of the features differently than training data like match week.

For match week, we use the dates of matches and define game weeks according to the fixtures on the official website. We parse our Date variable as a Date Time object for the same. It was observed that some match which was supposed to be played in match week 3 was postponed and later played after several scheduled match weeks. In such cases, we have hardcoded the dates and match weeks. We have also calculated the Goals scored, Goals conceded, the cumulative points, the form, and the form points (considering the previous five matches played by each team). We have also scaled the points scored by the team according to match week as it tends to increase with match weeks. We have also dropped testing data related to the team 'Brentford' as it is a new team in the league with no historical data.

Our training data included 380 matches, i.e., 38 match weeks for each season of the league; however, as the league considered for testing is undergoing, the number of match weeks is less. Thus, for all features dependent on match week, we have kept the matches not played till now as null and denoted them often using X. The testing data is independent of the playing ground and is grouped by each team and match week. The team names are one-hot encoded, and the data is then fed to our model.

C. Modelling

For modelling, we will use the training data and testing data as prepared in the above sections. Our

independent variables include all the features in the testing data, and our dependent variable is 'Win', a categorical variable created by us. For validation, we would be splitting our training data into train and validation sets with validation size being 25%, i.e., our training set has 8,261 rows, and validation has 2,754. Similar to the first part of the research, we have tested and evaluated nine state-of-the-art classification algorithms, as mentioned in Fig 2. The accuracy and cross-validation scores of the models are also given for comparison in Table IV.

TABLE IV
MODEL EVALUATION RESULTS

Model Name	Accuracy Score	Cross Val Score
XBGClassifier	0.946	0.978
Logistic Regression	0.95	0.974
RandomForestClassifier	0.952	0.976
DecisionTreeClassifier	0.937	0.973
KNeighborsClassifier	0.959	0.966
SVC	0.949	0.97
GaussianNB	0.692	0.505
AdaBoostClassifier	0.948	0.979
GradientBoostingClassifier	0.945	0.977

According to accuracy and cross-validation score, the AdaBoost classifier seemed to fit our dataset perfectly. In this situation, we did not see a need to use a fusion model as in 'Prediction of Football Match Results Based on Model Fusion' [8] by Quan Zhang, HongZhen Xu, Li Wei and LiangQi Zhou. Thus, we use AdaBoost for our testing data and predict probability function. This function gives the probability of a team winning or losing the league at the end of each match week. We represent the probability in terms of a percentage up to 2 decimal places.

D. Results

Post modelling, we are getting an output that includes the team's name, match week, and the probability of winning the league based on performance that week. We wished to have interactive user input and visual representation of the result. For this purpose, we developed a simple application using Flask where we can take two inputs from the user: the team whose winning chances are to be predicted and the match week for which the winning chances of all teams need to be predicted as shown in Fig. 2. The output is then manipulated

according to the user input, and the results are presented in the form of an interactive chart using JSON [9], Plotly [10], and plotly express.

Week wise EPL winning probability

Fig. 2 User Input Interface

1) *Team Wise League Winner Prediction*: This part shows the trend of winning probability using a line plot. The current winning probability for the team (Chelsea in this case) is also reported, as shown in Fig. 3.

Winning probability of Chelsea across Matchweeks

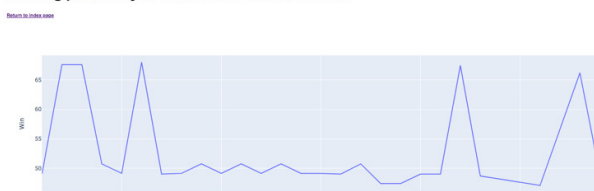


Fig. 3 Team Wise Results

2) *Match Week Wise League Winner Prediction*: The line plot shows each team's probability of winning the league according to their performance in that week (20 in this case) and historical data. This is a strong tool for comparison, and the team with the highest chances is also reported, as shown in Fig. 4.

Winning probability of teams in Matchweek 20



Fig. 4 Match Week Wise Results

V. CONCLUSIONS

The research extends the work done previously by authors in 'Full-Time Result Prediction using Ensemble Techniques' [1]. The first part of the research predicted a match-winner between any two teams in the English Premier League 2021-22. The

work required only inputting the basic features of the playing teams, and every important metric was calculated automatically before feeding the data to the model. Adding features like form and form points and the cumulative sum of previously present features enhanced the accuracy by 5%, i.e., 80% on the validation set. The comparison graph is shown in Fig. 5. The same dataset was used for the second part of the research.

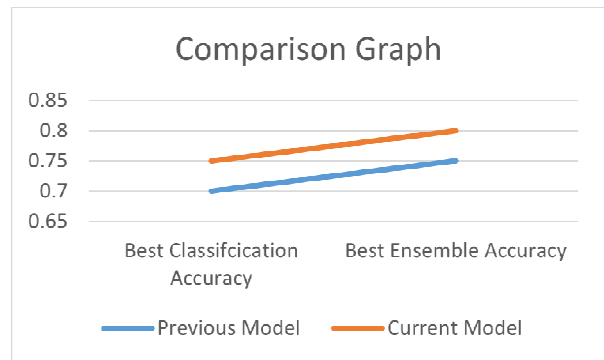


Fig. 5 Comparison of model in [1] with current model

This part's training and test set was created at the match week level instead of the match level as in part one. The goal of this part was to predict the league winner at the end of each playing week. The model was trained using current and historical team playing statistics at a match week level. The model resulted in 94.8% accuracy. The model was then tested on data of EPL 2021-22, and the results were visualized. The entire research work is done using the data of the English Premier League but can be easily extended to any football league.

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